

C# Console Application Assignment

Word Guessing Game

Problem Statement:

The goal of this project is to build a console-based word guessing game like Wordle using C# and Object-Oriented Programming concepts.

The system will randomly choose a hidden 5-letter word from a predefined collection. The player gets a maximum of 6 attempts to guess the correct word.

After every guess, the system should provide feedback for each letter:

G → correct letter in correct position

Y → correct letter but wrong position

X → letter not present in the hidden word

The game must also validate user input properly using custom exception handling.

Additional features like replay option, score tracking, difficulty levels, duplicate guess checking, and colored console output can also be included.

Core Logic for the game (Pseudocode):

1. Click on Start Game
 - a. Random hiddenWord selected
 - b. Initialize attemptCount = 0 -> when user starts to play
 - c. attemptCount++
2. User starts guessing the 5-letter word
 - a. userInput -> stored when user enters the 5 letter word
 - b. check if (userInput) : (user input validation)
 - i. empty input
 - ii. less than 5 characters
 - iii. greater than 5 characters
 - iv. numbers present
 - v. special characters present
 - vi. duplicate guess
 - c. else -> exception handling for invalid cases

- i. throw custom exception
 - ii. display proper message
 - iii. but attemptCount for this should not reduce
- 3. Feedback Generation
 - a. For each char in userInput -> validate with hiddenWord
 - i. Same letter + same position → G
 - ii. Letter exists elsewhere → Y
 - iii. Letter not found → X
- 4. Win Condition
 - a. If all 5 letters position == 5 ?
 - b. game ends immediately
 - c. prompt win message
- 5. Lose Condition
 - a. attemptCount = 6?
 - b. reveal hidden word
 - c. prompt “better luck next time”
- 6. Replay Option
 - a. Prompt user to play again

Project folder structure:

```
WordleGame/  
|  
├── Models/  
|   Attempt.cs  
|  
├── Core/  
|   Game.cs  
|   WordProvider.cs  
|   GuessValidator.cs  
|   FeedbackProvider.cs  
|  
└── Exceptions/
```

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C# Gam

```
| InvalidGuessException.cs  
|  
└─ Program.cs
```

Bonus Requirements

1. Color output
2. Replay option
3. Score system
4. Difficulty levels

Link to the GitHub Repo - [Click on the file](#)